

LECTURE 7:

Cooperation in MAS (III):

Negotiation via auctions

Introduction to Multi-Agent Systems (MESIIA, MIA)

Department of Computer Science and Mathematics

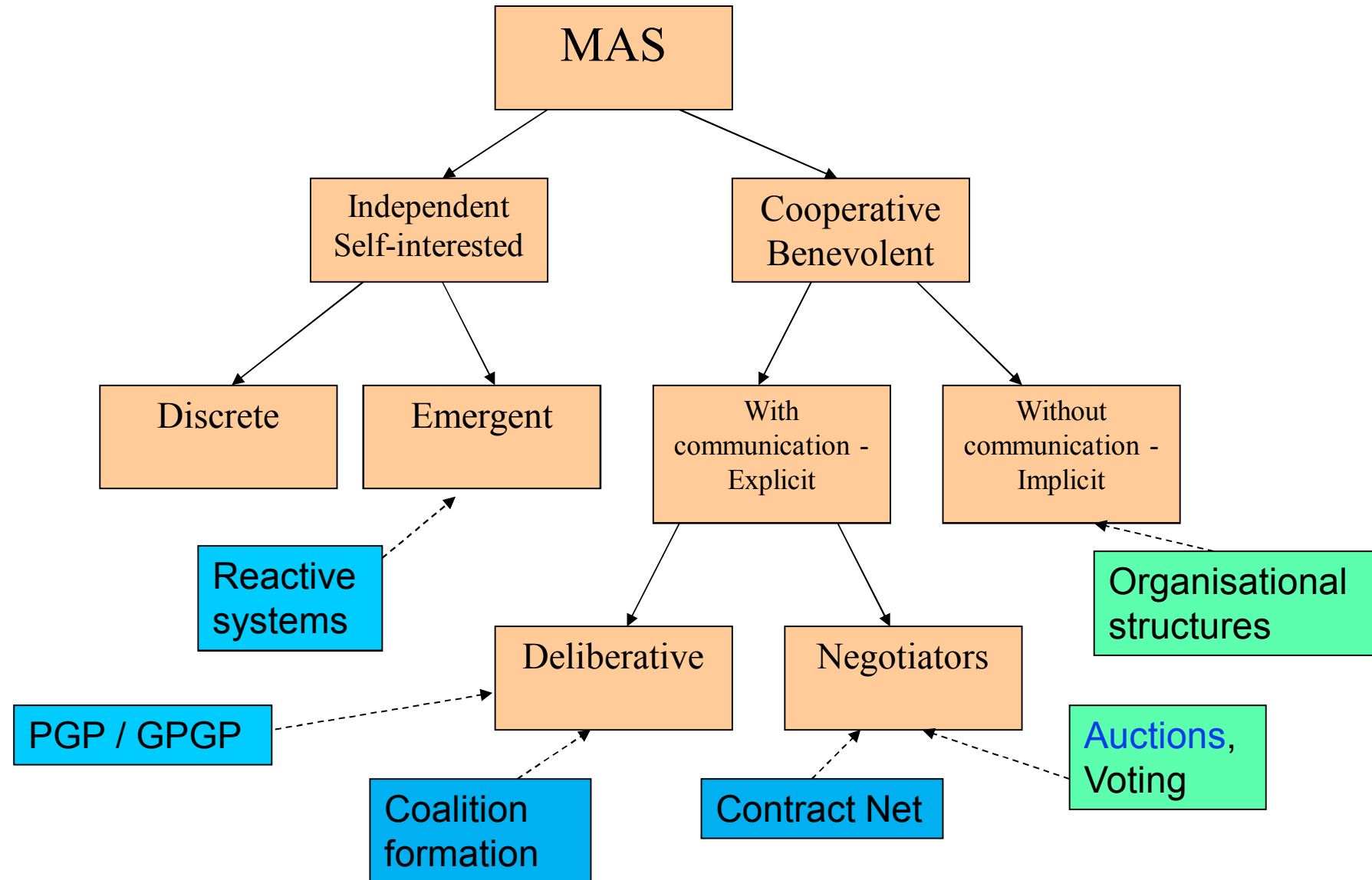
Rovira i Virgili University

Fall 2019

Lecture overview

- Negotiation protocols
 - Auctions
 - English/Dutch/FPSB/Vickrey
 - Multi-unit
 - Multi-attribute
 - Combinatorial
-

Cooperation hierarchy [last lectures]



Negotiation between multiple agents



Need of negotiation in MAS

- Agents may have incompatible goals, and the resources to achieve these goals may be limited; in such cases **competition** and **conflicts** may arise
 - The effects of the agents' antagonistic behaviour need to be limited
 - Agents must be able to **reach compromises**, **resolve conflicts**, **allocate goods and resources** by way of an **agreement**
 - Agents' interactions are governed by a set of rules: an **interaction protocol**
-

Negotiation protocol elements (I)

Private elements

- The set of **strategies** that the agents can use to participate in the negotiation process:
 - They are not dictated by the protocol itself
 - They may take into account the strategies of the other agents
 - Learning mechanisms
-

Negotiation protocol elements (II)

Public elements

- A **negotiation set** which represents the space of possible offers/proposals that the agents can make
 - The **protocol rules** which govern the agents' interactions
-

Protocol rules (I)

- Admission rules

- When an agent can participate in a negotiation (e.g. eligibility criteria)

- Interaction rules

- Sequence of admissible/valid actions (e.g. moments in which bids are allowed)

- Validity rules

- What constitutes a legal offer/proposal (e.g. a new bid must be higher than the last bid)

- Outcome determination rules

- When an agreement has been reached
-

Protocol rules (II)

- Withdrawal rules

- When an agent can withdraw from the negotiation

- Termination rules

- When a negotiation ends unsuccessfully

- Commitment rules

- How the commitments that agents make during the negotiation are managed
-

Negotiation factors

- **Number of attributes:** one, many
[Multi-attribute auctions]
 - **Number of agents:**
 - One-to-one
 - One-to-many
 - Many-to-many
 - **Number of units:** one, many
[Multi-unit auction]
 - **Interrelated goods:** one good or a number of goods that are substitutable or interdependent
[Combinatorial auctions]
-

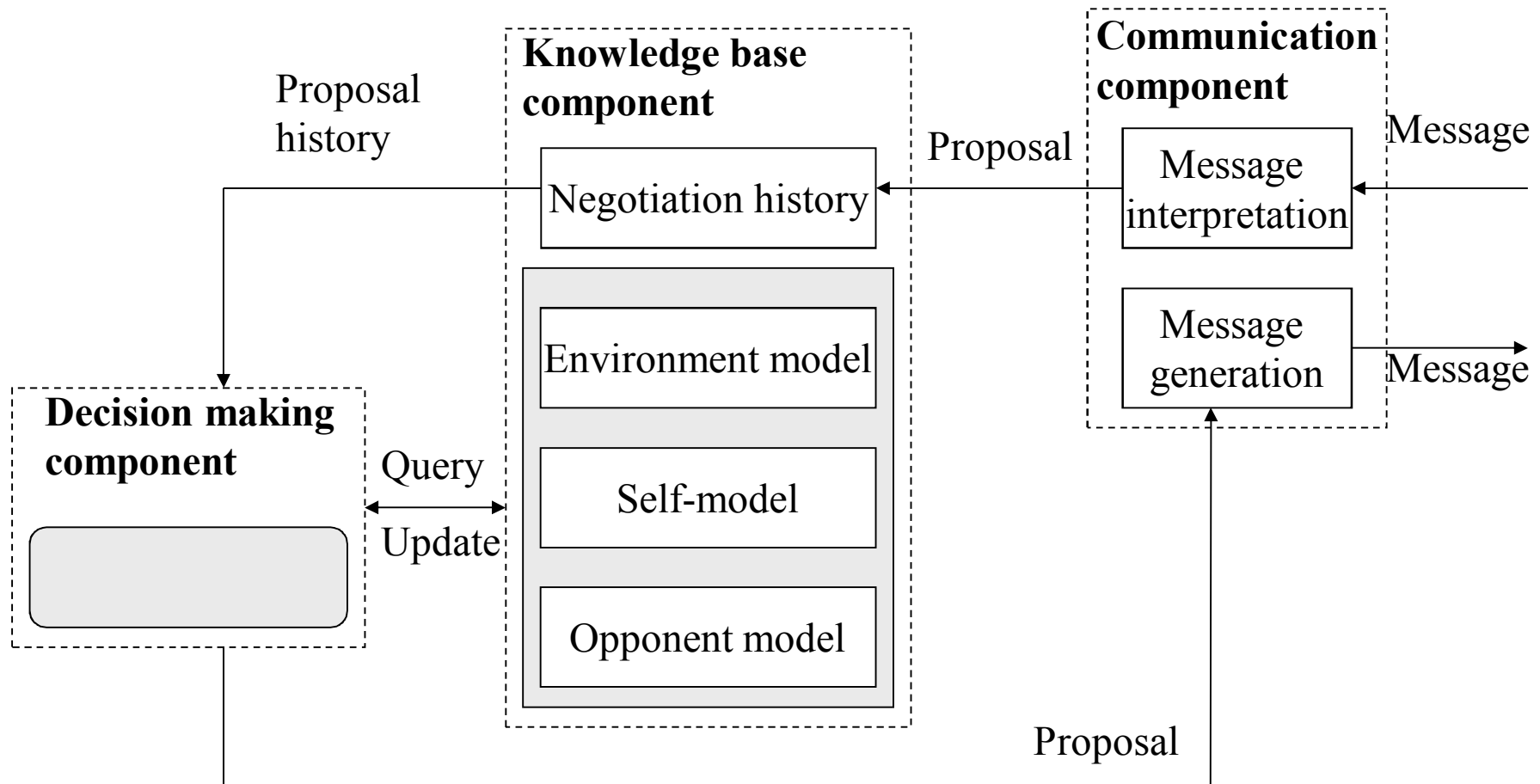
Protocol evaluation criteria (I)

- **Social welfare** – the sum of all agent's payoffs or utilities in a given solution
 - **Pareto efficiency** – a solution x is **Pareto optimal** if there's no other solution x' such that at least one agent is better off in x' than in x , and no agent is worse off in x' than in x
 - **Individual rationality** – an agent should not lose out by participating in a negotiation
-

Protocol evaluation criteria (II)

- **Stability** – mechanism should be designed to be non-manipulable: motivate each agent to behave in the desired manner
 - **Computational efficiency** – mechanisms should be designed so that when agents use them, as little computation is needed as possible
 - **Distribution and communication efficiency** – distributed protocol vs. minimum communication (time, money,...)
-

Architecture for negotiating agents



Auctions

- A class of negotiation protocols which provide us with methods for allocating goods/resources based upon **competition among self-interested parties**

Where are auctions used?

- Telecommunication and TV licenses, mining rights, emission permits, airport gates and takeoff/landing slots
- Google AdSense **[article in Moodle space]**
- Extremely popular as a means for conducting consumer-to-consumer (C2C) negotiations online (eBay, Yahoo)
- Collectibles (paintings, books, antiques)
- Agricultural products (flowers, fish, tobacco)

Advantages of auctions

- Markets may not exist for what the seller wants to sell (phone spectrum)
 - The seller does not know how much an item is worth
 - Create competition; enhances the seller's bargaining power
 - Flexibility
 - Less time-consuming and expensive than negotiating a price
-

Auction participants

The auctioneer

- ❑ A seller who wants to sell goods at the highest possible price
- ❑ Someone who wants to subcontract out contracts at the lowest possible price
- ❑ A buyer who wants to buy a good

The bidders

- ❑ Buyers who want to acquire goods at the lowest possible price
 - ❑ Contractors who want to get a contract at the highest price
 - ❑ Sellers who want to sell their goods
-

Properties of participants

- Agents are supposed to be:
 - **Self-interested**: they are seeking to maximise their own payoffs
 - **Rational**: they always prefer a larger payoff than a smaller one
-

Basic concepts (I)



- **Bids:** offers by bidders to **buy** or **sell** the auctioned item
 - **Buy bid:** the price a bidder is willing to pay to own an item
 - **Sell bid:** the price a bidder is willing to accept to sell an item
- **Reservation price:** Maximum/minimum price that a buyer/seller is willing to pay/accept for an item – usually private information

Basic concepts (II)

- **Process a bid**: the auctioneer checks the validity of a bid according to the rules of the auction protocol and updates its database
 - **Clearance**: matching between buyers and sellers and setting of the transaction price
 - **Clearing price**: final transaction price that the winning bidder pays and the auctioneer receives
-

Typical auction process (I)

- Buyers and sellers **register** with the auction house
 - The auction event is **set up**
 - The auction is **scheduled and advertised** (local press, Internet, etc.)
 - The actual **bidding phase** takes place according to the rules of the auction
 - **Information may be revealed** to participants to some extent during the bidding phase
-

Typical auction process (II)

- The **bids are evaluated** and the auction closes with a winner being determined if any of the bids is successful
 - The **transaction phase** takes place: the buyer pays for the goods and the seller ships/delivers them
-

Classification of auctions (I)

(1) **Bidding rules**

- Single good or combinatorial
 - Single attribute or multi-attribute
 - Open (outcry) or sealed-bid
 - Ascending or descending
 - Single unit or multi-unit
-

Classification of auctions (II)

(2) Information revelation policy

- **When to reveal information:** on each bid, at predetermined points in time, on inactivity, etc.
 - **What information**
 - Bid: the price a seller would have to offer in order to trade
 - Ask: the price a buyer would have to offer in order to trade
 - Auction closure: known, unknown, after a period of inactivity
 - **To whom:** participants only, everyone
-

Classification of auctions (III)

(3) Clearing policy

- **When to clear:** on each bid, on closure, periodically, after a period of inactivity
 - **Who gets what** (allocation and winner determination problem)
 - **At what price:** first, second price or other
-

Auction formats

- Basic
 - English
 - Dutch
 - FPSB-First price sealed bid
 - Vickrey
 - Multi-unit
 - Multi-attribute
 - Combinatorial
 - Others ...
-

English auction (I)

- Open-outcry and ascending-price auction
 - The auctioneer announces an opening price or the reserve price
 - Bidders raise their bids and the auction proceeds to successively higher bids
 - The winner of the auction is the bidder of the highest bid
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English auction (II)

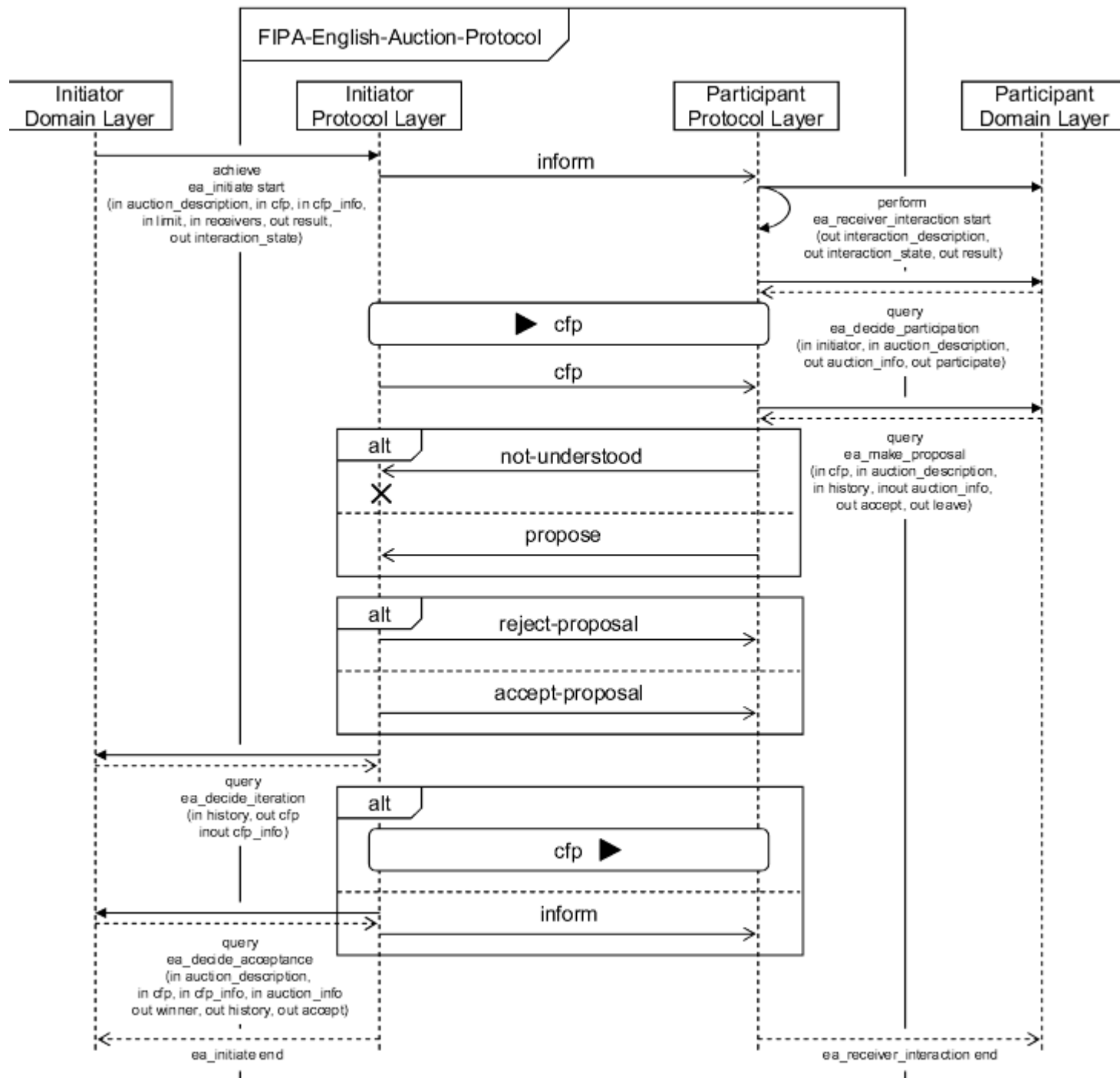
- The best strategy is to **bid a small amount above the previous bid** until one reaches its private value and then stop
- The bidders gain information by *observing* the others' bids
 - Amount of bid increase, doubts, time taken to bid, ...

English auction disadvantages (I)

- The **reserve price may not be met** and the item may remain unsold
 - The **auctioneer may cheat** by overstating a reserve price or present a reserve price that does not exist
 - Susceptible to **rings** (collusions)
 - **Phantom bid**
 - The auctioneer calls a bid that no one has made
-

English auction disadvantages (II)

- Bidders can become carried away and overbid
 - Vulnerable to shills
 - Person associated to the seller that pretends to be an enthusiastic customer
 - In real life it can become complicated (voices, signals)
-



Dutch auction (I)

- Open and descending-price auction
 - The auctioneer announces a very high opening bid
 - Then the auctioneer keeps lowering the price until a bidder accepts it – the winner pays the price of its bid
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Fish
auction
in Palma

Fish
auction
in
Montgat







[Video](#)

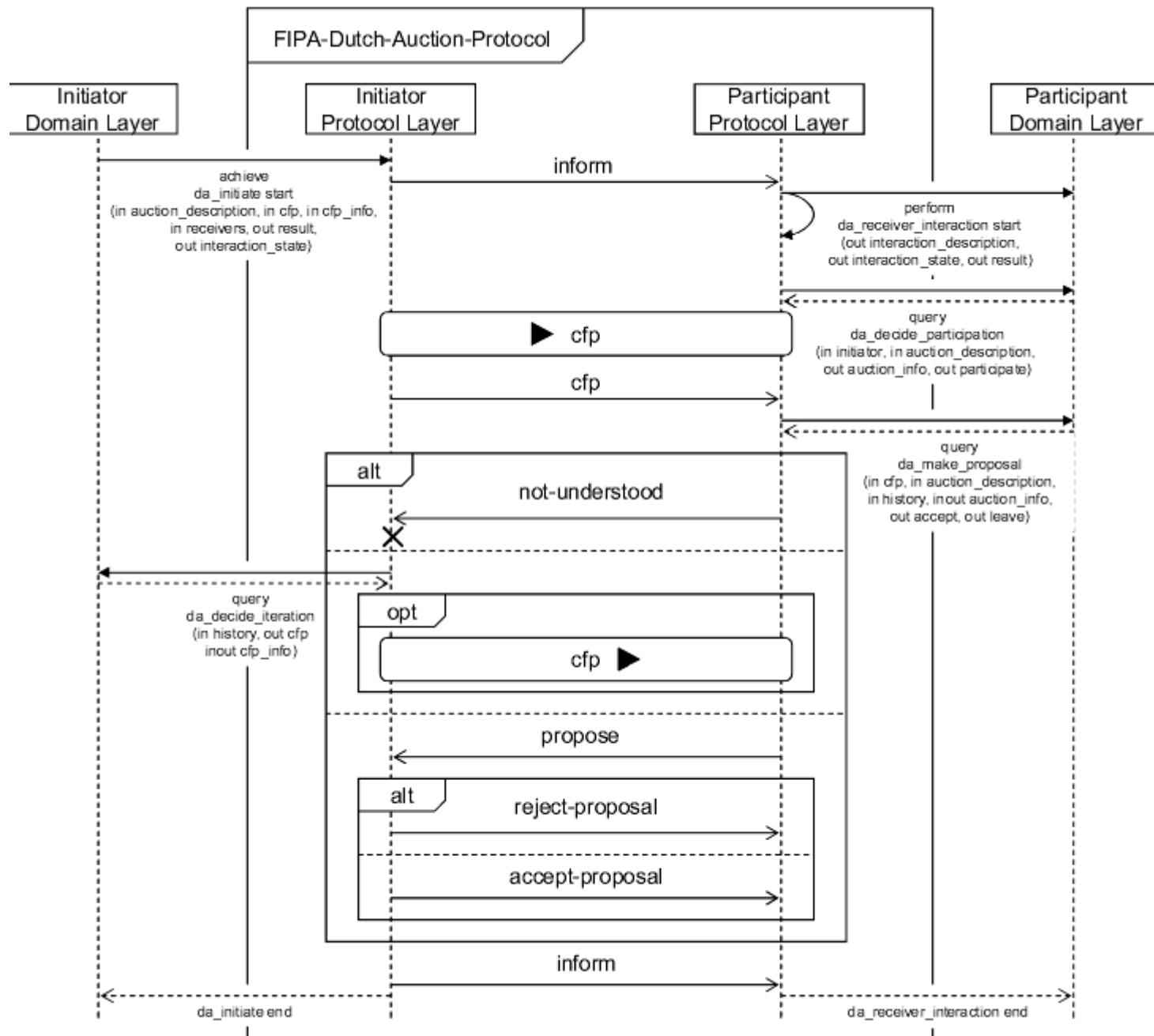
[More detailed video](#)

Dutch auction (II)

- Bidders need to decide in advance the maximum amount that they are willing to bid
 - A bidder must decide when to stop the auction (bid) based upon its own valuation of the commodity and its prior beliefs about the valuations of the other bidders
 - No relevant information on the valuation of the other bidders is disclosed during the process of the auction until it is too late
-

Comments on Dutch auctions

- Very efficient in real time: used in The Netherlands for selling fresh flowers, Ontario tobacco auction, fresh produce auctions, fish markets
 - The effect of competition on the buyers is stronger in the Dutch auction than in the English one
-



First-price sealed-bid auction (I)

- Each bidder submits its own bid (usually in writing) without knowledge of the bids of others
 - Two phases:
 - (i) the **bidding phase** in which participants submit their bids
 - (ii) the **resolution phase** in which the bids are opened and the winner is determined
 - The highest bidder wins and pays the amount of its bid
-



First-price sealed-bid auction (II)

- An agent's strategy depends on its own valuation and prior beliefs about the other bidders' valuations
 - A higher bid raises the probability of winning, but lowers the bidder's profit if it wins
 - Bidders are better off not bidding their true valuations but a **small amount below it**
 - Trade-off between the profit they can get and the probability of losing the auction
-

FPSB Multi-unit auctions

- In many kinds of auctions there is a certain quantity of units of the same item, which may be distributed between different buyers
 - In a **multi-unit auction** sealed bids are sorted from high to low, and the items are awarded at the highest bid price until the supply is exhausted
-

Multi-unit auction example – 1000 units

- 200 units at 35 euros
- 250 units at 30 euros
- 300 units at 28 euros
- 100 units at 27 euros
- 150 units at 26 euros
- 500 units at 25 euros
- 225 units at 20 euros
- ...



Awarded units
(mean price:
29,5 euros/unit)

Vickrey auction

- A **second-price sealed-bid auction**, also known as *uniform second-price sealed-bid* or the *philatelist auction*
- Two distinct phases: the **bidding phase** and the **resolution phase**
- The highest bid wins but the bidder pays the amount of the **second-highest bid**



William
Vickrey,
Economics
Nobel Prize
1996

Why do Vickrey auctions work?

- Although it seems that the auctioneer would make more money out of a first-price sealed auction, this is *not* the case
 - Bidders adjust their bids upwards since they are not deterred by fear that they will have to pay too much
 - The price that the winning bidder pays depends on the others' bids alone and not on any action that the bidder undertakes
-

Comments on Vickrey auctions

- The best strategy is for the agent to *bid its true valuation*
 - The bidders do not waste time in *counter-speculating* what the other bidders will do
-

Best strategy (I)

A bidder has a dominant strategy to bid an amount equal to their valuation, that is: $b_i = v_i$

- Consider bidding $v - x$ if your true value is v
 - Suppose the highest bid other than yours is w
 - **Case 1:** If $v - x > w$ you win the auction and pay w , just as if you had bid v
-

Best strategy (II)

- **Case 2:** If $w > v$ you lose the auction and get nothing, just as if you had bid v
 - **Case 3:** If $v > w > v - x$ bidding $v - x$ causes you to lose the auction, whereas if you had bid v , you would have won the auction and paid w for a net surplus of $v - w$
 - Hence you can only lose, and never gain, by bidding $v - x$
-

Combination: “Anglo-Dutch” auction

- Auctioneer runs an **English auction** in which the price is raised continuously until all but two (or n) bidders have dropped out
 - The remaining bidders are then required to make a **final sealed-bid offer** that is no lower than the current asking price, and the winner pays the bid
 - Sealed-bid at the **final stage** induces some uncertainty about which of the finalists will win, and entrants are attracted by the chance to make it to the final
-

Multi-attribute auctions

- *Multi-attribute* or multi-dimensional auctions allow bidders to **submit bids on more than one attribute** or dimension of a good
 - Very common in procurement situations
 - The attributes under negotiation are usually **defined in advance** and bidders can compete in open-cry or sealed-bid auctions
-

Example: a procurement auction (I)

- Consider a manufacturer which uses raw materials from a number of suppliers to produce finished goods
 - Assume the manufacturer requires 1000 units of A by March 31st
 - The manufacturer sends a *Request For Quotes* (RFQ) to all potential suppliers of A
-

Example: a procurement auction (II)

- Suppliers reply with a bid b_i
 <quantity, delivery_date, price>
 - The *delivery_date* and the *quantity* may deviate from those requested by the manufacturer
 - The *price* requested by the various suppliers will differ too
 - The manufacturer's request could be covered by one or more bids in combination (sole or multiple sourcing)
-

Winner determination

- A *multi-attribute offer* has the form $\mathbf{v}_i = (v_{i1}, \dots, v_{ij})$ where v_{ij} is the level of attribute j offered by i
- Each bid is evaluated through a scoring function $S(\mathbf{v}_i)$ which is a **weighted average** function – often scaled from 0 to 1

$$s_i = S(\mathbf{v}_i) = \sum_{j \in J} w_j S_j(v_{ij}) \text{ and } \sum_{j \in J} w_j = 1$$

Example: 1000 units needed by 31/03

S1: 500 units, 28/03, 30 euros/unit

S2: 1000 units, 4/04, 38 euros/unit

S3: 700 units, 2/04, 37 euros/unit

S4: 300 units, 30/03, 42 euros/unit

- If completeness is preferred => S3+S4 or S2
 - S2 (1000 units, 38000 euros, by April 4th)
 - S3+S4 (1000 units, 38500 euros, by April 2nd)
 - If sole-sourcing is preferred => S2
 - S2 (1000 units, 38000 euros, by April 4th)
 - If deadline has to be kept => S1+S4
 - S1+S4 (800 units, 27600 euros, by March 30th)
-

Advantages of multi-attribute auctions

- They allow **more degrees of freedom** for bidders: price may not be the only attribute of interest
 - **More efficient information exchange** among the market participants
-

Combinatorial auctions

- Auctioning multiple *distinguishable* items when bidders have preferences over combinations of items: *complementarity* & *substitutability*

Example: auctioning a **three-seater sofa**, a **two-seater one, two armchairs** and a **coffee table**

- What's the best way to sell these goods? As a complete set? Each item separately? Allow combinations?



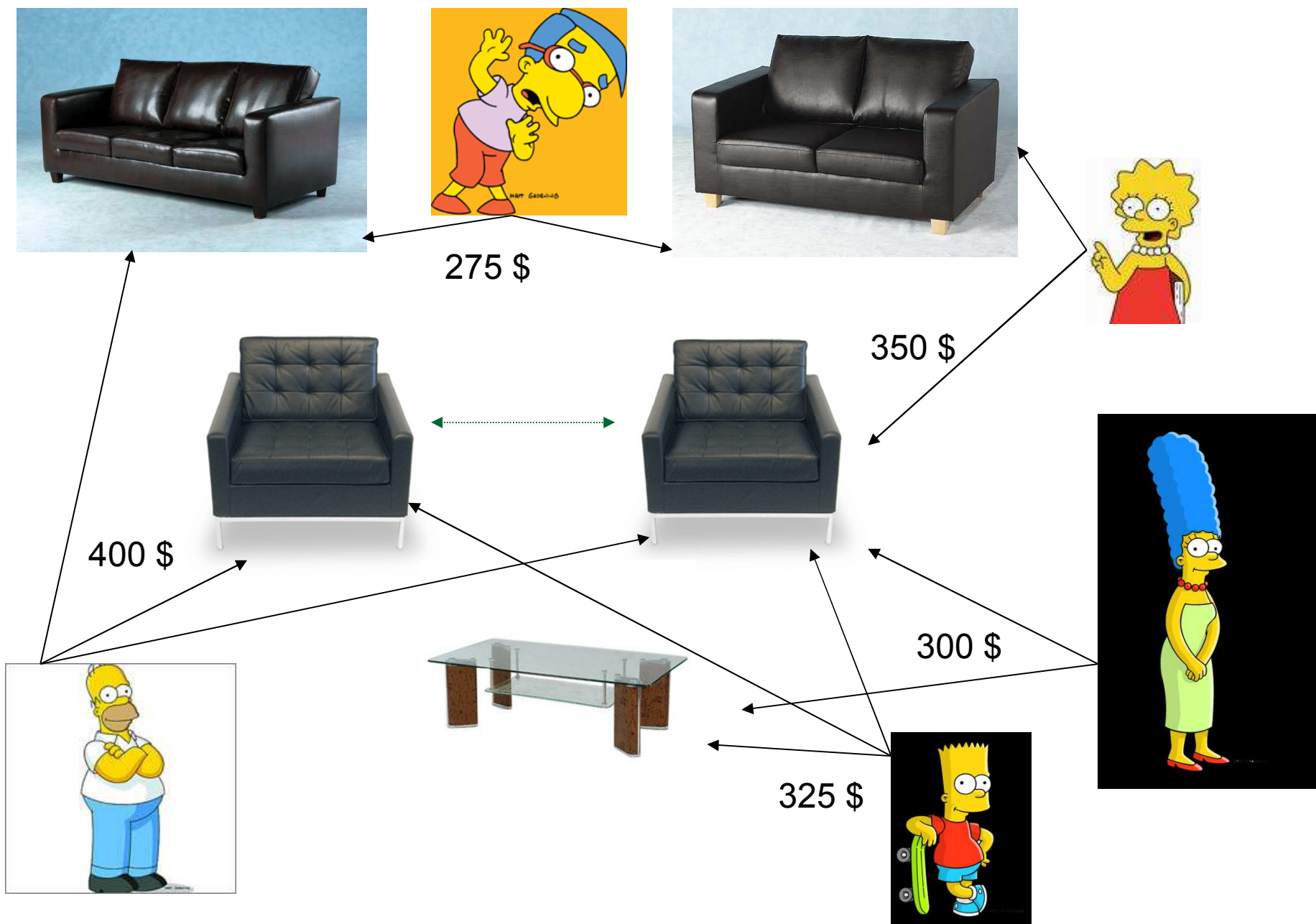
Sequential vs parallel auctions

- **Sequential auctions:** run individual auctions one after the other
 - ❑ Impossible to determine best strategy because game tree is huge
 - ❑ Inefficiencies can result from future uncertainties

 - **Parallel auctions:** run individual auctions in parallel
 - ❑ Inefficiencies can still result from future uncertainties
 - ❑ Difficult to keep track of several simultaneous auctions with substitutable and interdependent goods (at least for human agents)
-

Bundles of items

- *Combinatorial* auctions: bids can be submitted on combinations (bundles) of items
 - Example
 - Bundle1: the two-seater sofa and the table
 - Bundle2: the armchairs and the table
 - Bundle3: the two-seater sofa and the armchairs
-



Interest of combinatorial auctions

- Bidder's perspective

- Avoids the need for lookahead
- Optimal bundling of items

- Auctioneer's perspective

- Winner determination problem: Decide on the winning bids so as to maximize the sum of bid prices (i.e. revenue)



Issues on combinatorial auctions

- Should the combinations of items on which bids are allowed be restricted?
 - Limits number of possibilities to be considered
 - Single-shot or iterative combinatorial auctions?
 - In an iterative auction, what information should be made available to bidders from round to round?
 - What is the objective of the auctioneer?
 - Sell all items
 - Maximise revenue
 - Minimise number of buyers
-

Advantages of auctions

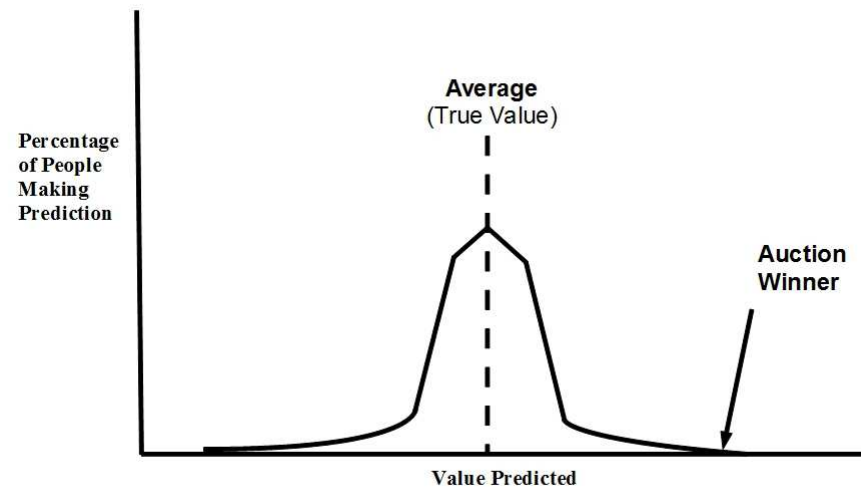
- **Flexibility**, as protocols can be tailor-made
 - **Less time-consuming and expensive** than negotiating a price
 - **Simplicity** in determining the market prices
-

Disadvantages of auctions

- Winner's curse
 - Lying auctioneer
 - Sniping
 - Others: rings, revelation of private information, ...
-

Winner's curse

- What bidders suffer when they win an auction by **overestimating** how much something is worth and therefore **bidding too much** – the most optimistic bidder wins
- *A win may actually mean a loss ...*



Facing the winner's curse

- Anticipate the effects of the winner's curse before bidding:
 - ❑ Assume that you have the **highest estimate**
 - ❑ Rational bidding: correct **downwards**
 - Any piece of information is useful to the bidder – **beware of rings !!!**
 - ❑ How aggressively others bid
 - ❑ How many remain in the bidding
 - ❑ When others apparently drop off the bidding
-

Lying auctioneer

- In the Vickrey auction, the auctioneer may **overstate the second-highest bid**
 - ❑ Solution: use of cryptographic electronic signatures
 - ❑ It cannot happen in the other 3 protocols
- In English auctions: **shills**
[Read NYTimes article]
- In sealed-bid auctions: **Auctioneer may place a bid himself** (reservation price)



Sniping

Sniping: bidding very late in the auction in the hope that other bidders do not have time to respond and you can snatch a bargain. This is an issue, in particular in online auctions

Bidders may bid late in order to avoid:

- increasing the price of the item early on in the auction
 - revealing their preferences early in the auction (especially experts)
 - bidding wars with other like-minded agents
-

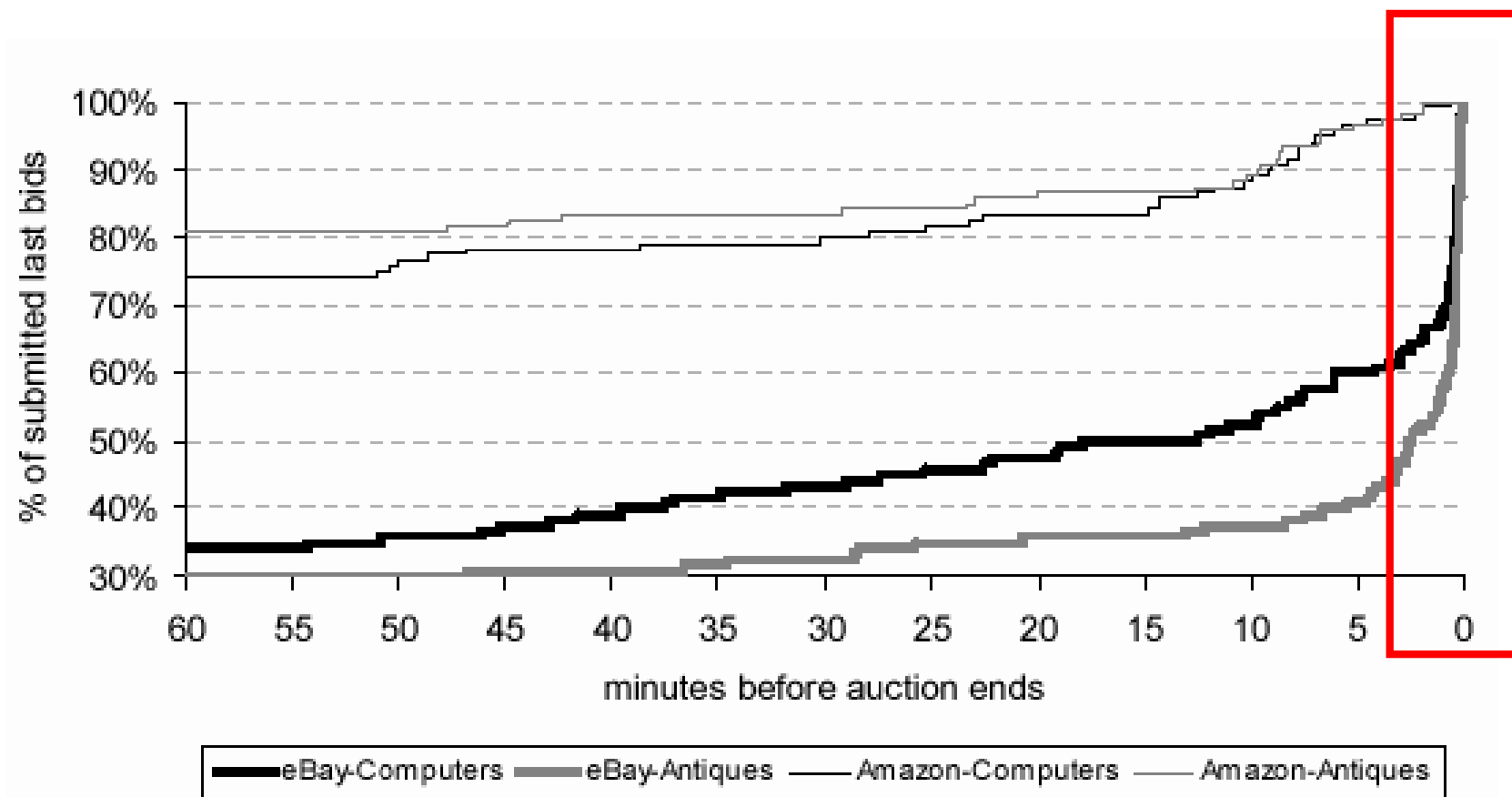


Figure 1b—Cumulative distributions over time of auctions' last bids



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"Powersnipe pays for itself after the first use." - J. Thomas, PhD, eBay Guru

"Saves you time and improves your odds of victory." - Time Magazine

What is PowerSnipe Auction Sniper?

PowerSnipe is an eBay Auction Sniper software and service that bids for you automatically on eBay in the closing seconds of the auction. eBay auctions are often lost in the last few seconds of the auction because the majority of people wait until the last seconds to place their bids. PowerSnipe securely bids for you in the final moments. This gives you a better chance of winning and helps you to avoid getting into "bidding wars" with other users competing over an item. Set it and forget it. Your computer does not have to be turned on at the end of the auction.

How do I get Powersnipe?

To get Powersnipe [Sign Up](#). Users can access Powersnipe services through the [software](#), or through the website.

Why is PowerSnipe Necessary?

When an eBay auction is placed it usually does not end for days. One person places his maximum bid. Later another person comes along and bids until he is the highest bidder. During the course of an auction this process often repeats itself many times, it is known as a "bidding war" and it drives the price of an auction up.

To avoid the "bidding war" people have started bidding another way known as "snipe" or "sniping". They wait until the final

What are the Benefits?

- Places your bid just before auction ends.
- Pay once and get unlimited usage.
- Your computer doesn't have to be on.
- Works with all eBay sites.
- Inexpensive and Guaranteed.
- Daily auto search delivers eBay search results to your email every day.
- Bid Groups allow you to bid on the same item from different sellers until you win at the price you set.
- Fast, award-winning customer support.

How does it work?



PowerSnipe eliminates all of the problems associated with bidding and gives you the best possible chance of winning the auction. PowerSnipe can be used on any computer connected to the Internet. When you are ready to place a bid you can either copy and paste the auction number into your PowerSnipe account, or use the software [eBay Auction Sniper and Auto Search](#) and press the snipe button.

PowerSnipe will extract all the information from the auction page and save it in an easy to review format.

Online auction systems

- **Online auctions** are increasingly popular: last minute package holidays, C2C negotiations (eBay), B2B negotiations
 - Popularity and success of online auctions can be attributed to:
 - **Simple mechanisms** used
 - Allow **millions of people** with no specialist knowledge of negotiation protocols and strategies to exchange goods
-

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[Advanced Search](#)Hello! [Sign in](#) or [register](#).[Back to item description](#)**Bid History**Item number: [150103980058](#)[Email to a friend](#) | [Watch this item](#) in My eBay

i As the internet evolves, eBay continues to strike a balance between preserving transparency and protecting our Community of members. eBay has decided to change how bid history information is displayed so bad guys cannot target bidders with fake offers using this information. In certain cases, some bidders will no longer be able to view Bidder User IDs on the Bid History page. Your User ID will be shown only to you and the seller of the item you're bidding on. Other members will see an anonymous name, such as Bidder 1, applied consistently to the Bid History page.

Item title: PS3+xbox 360+ipod+Tivo+36 games NO Reserve!!! [\(revised\)](#)

Time left: 1 day, 16 hours 2 minutes 58 seconds

Only actual bids (not automatic bids generated up to a bidder's maximum) are shown. Automatic bids may be placed days or hours before a listing ends. [Learn more about bidding.](#)

Bidder ?	Bid Amount	Date of bid	Bidding Details
Bidder 10 ?	US \$600.00	Mar-21-07 22:00:02 PDT	View
Bidder 1 ?	US \$590.00	Mar-22-07 03:21:56 PDT	View
Bidder 1 ?	US \$570.00	Mar-22-07 03:21:29 PDT	View
Bidder 9 ?	US \$550.00	Mar-21-07 20:52:39 PDT	View
Bidder 8	US \$525.00	Mar-21-07 19:43:43 PDT	View
Bidder 7	US \$510.00	Mar-21-07 09:29:50 PDT	View
Bidder 1 ?	US \$500.00	Mar-20-07 23:01:04 PDT	View
Bidder 6 ?	US \$500.00	Mar-21-07 09:05:25 PDT	View
Bidder 1 ?	US \$400.00	Mar-20-07 02:49:23 PDT	View
Bidder 6 ?	US \$400.00	Mar-20-07 22:27:14 PDT	View
Bidder 5	US \$330.00	Mar-20-07 20:40:03 PDT	View
Bidder 4	US \$300.00	Mar-20-07 08:44:09 PDT	View
Bidder 4	US \$200.00	Mar-20-07 08:43:57 PDT	View
Bidder 4	US \$150.00	Mar-20-07 08:43:41 PDT	View
Bidder 3	US \$120.00	Mar-20-07 07:01:04 PDT	View
Bidder 3	US \$100.00	Mar-20-07 07:00:49 PDT	View
Bidder 3	US \$90.00	Mar-20-07 07:00:40 PDT	View
Bidder 3	US \$83.00	Mar-20-07 07:00:30 PDT	View
Bidder 2 ?	US \$80.00	Mar-20-07 04:33:41 PDT	View
Bidder 2 ?	US \$50.00	Mar-20-07 04:33:28 PDT	View

Pitfalls in auction design

- Collusive behaviour
 - Entry deterrence
 - Setting reservation prices at the right level
 - Communication and computational efficiency
 - Advanced auction formats may be cognitively complex and difficult to understand by participants
-

Use in practical exercise

- The coordinator could make an **auction** for each example to predict to determine which agent(s) should answer it
- The bids of the intelligent agents could depend on their current state (idle, the current accuracy of the agent, ...)

Readings for this week

- Chapter 8 in M.Fasli's book (*Agent technology for e-commerce*)
- New York Times article (June 2000): *In online auctions, rings of bidders*
- Chapter 14 in M.Wooldridge's book (*An Introduction to MultiAgent Systems*), 2nd edition



